

GUNJIT VALECHHA

Data Engineer | Python | SQL | PySpark | Apache Spark | Azure Databricks | Airflow

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EXPERIENCE

HDFC Bank — Data Engineer

Lakehouse Platform — Python, SQL, Apache Airflow, Trino

July 2023 – Present
Silver Star Award (2026)

- Enhanced the existing Lakehouse framework to develop the bank's in-house **Master Data Management platform**, removing unnecessary functionality and implementing custom capabilities for dynamic metadata insertion and automated creation of master and reference data tables, eliminating dependency on IBM's MDM solution.
- Designed MDM tables to store **~1 million reference and master-data records** and implemented ingestion validation by reconciling source row counts with target table counts and verifying the expected number of tables.
- Designed target Lakehouse schemas and data types for migration of dimensional data from **Netezza**, determining full vs. incremental load strategies based on source characteristics and refresh frequency.
- Implemented **SharePoint integration using Microsoft Graph API**, including token-based authentication and dynamic file retrieval logic to select the latest file or files from specified folders based on pipeline requirements.
- Extended existing ingestion pipelines with custom Excel parsing and transformation logic based on business requirements, leveraging **agentic AI coding tools** to reduce development time by **~60%** and rigorously testing the implementation ahead of schedule.
- Implemented **failure monitoring and email alerts** for Airflow DAGs, automatically notifying developers when pipeline execution failed.

Industry Analytics Platform — Python, PySpark, SQL, Selenium, Azure Databricks

- Delivered an in-house solution at INR 36L vs. INR 1.6–2.5 Cr external tools in 2 months, achieving **~85%** cost savings and enabling **~INR 1 Cr** in annual revenue through strategic advisory deliverables.
- Designed and deployed automated web scraping pipelines for **200+ heterogeneous data sources**, extracting structured data from HTML, PDF, Excel, CSV, and plain text formats into a centralized data store.
- Developed end-to-end PySpark ETL pipelines and Databricks workflows, creating **200+ data warehouse tables** used by policy analytics and corporate advisory teams.
- Managed production deployments via **Azure DevOps CI/CD** pipelines with automated testing gates, ensuring zero-downtime releases and consistent environment parity.

Loan Products Data Platform (Xpress Car, Personal & Business Loan) — Python, PySpark, SQL, Azure Databricks

- Owned end-to-end ELT pipelines for **100,000+ customer records** across Xpress Car, Personal, and Business Loan products, onboarding new source systems and developing target tables for data collected throughout the loan journey.
- Developed a parameterized **Azure Databricks notebook** to automate onboarding of new API source systems, generating metadata required for ingestion through a small set of input parameters and reducing pipeline development time by **~80%**.
- Optimized Databricks workflows by upgrading the Spark version, tuning cluster memory and autoscaling, and leveraging cluster pools to reduce pipeline runtime by **~20%** while controlling compute costs.
- Resolved production pipeline failures caused by source-data and schema changes, including datatype changes and unexpected input data, through root-cause analysis and documented runbooks to prevent recurrence.

SKILLS

Programming & Querying: Python, SQL, PySpark

Data Engineering: ETL/ELT, Data Pipeline Design, Data Modeling, Data Warehousing, Lakehouse Architecture, Data Quality, Data Governance

Big Data & Orchestration: Apache Spark, Apache Airflow, Azure Databricks, Trino

Cloud & DevOps: Microsoft Azure, Azure DevOps, Git, CI/CD

Databases & Backend: PostgreSQL, FastAPI, WebSockets, AsyncIO, Pandas

Computer Science: Object-Oriented Programming, Low-Level Design, Data Structures & Algorithms, DBMS

PROJECTS

Algorithmic Trading Dashboard [Active] | Python, FastAPI, PostgreSQL, React, Typescript, AsyncIO | GitHub

- Built a real-time market-data ingestion pipeline using **Angel One SmartAPI**, normalizing SNAP_QUOTE ticks and batching writes to PostgreSQL with duplicate protection; persisted LTP, OHLC, volume, open interest, and top-five order-book levels.
- Implemented an event-driven live dashboard using PostgreSQL **LISTEN/NOTIFY**, an asynchronous event loop, and WebSocket broadcasting to push database-backed market-data updates to connected browser clients.
- Developed **FastAPI** services and a React frontend for instrument discovery, historical OHLCV/tick analysis, volatility views, and ingestion-status monitoring.
- Containerized PostgreSQL, market-data ingestion, API, dashboard, scheduler, and frontend components using **Docker Compose** for reproducible local deployment.

EDUCATION

National Institute of Technology, Warangal

2019 – 2023

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming
- Awarded the **Institute Merit Scholarship** for outstanding academic performance